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OVERVIEW

I am an economist specializing in environmental economics and empirical industrial organization. My research examines how firms' pricing decisions and consumer choices shape the effectiveness of environmental incentives, with implications for policy design. My econometric research focuses on empirical Bayes methods and the estimation of structural models.

EDUCATION

University of Toronto , Toronto, Canada.	Expected 2027
PhD in Economics	
Fields: Environmental Economics, Industrial organization, Econometrics	
Committee: Victor Aguirregabiria (co-supervisor), Stephan Heblich (co-supervisor), Robert McMillan, Eduardo Souza-Rodrigues	
Sharif University of Technology , Tehran, Iran.	2021
MSc in Economics (Finance)	
Sharif University of Technology , Tehran, Iran.	2018
BSc in Computer Engineering	

WORKING PAPERS

1. **When Low-Carbon Incentives Do Not Reduce Emissions: Evidence from Brazil's RenovaBio** [Link]
Sole-authored; Job Market Paper; presented at TMEC 2026

PUBLICATIONS

1. **Empirical Bayes with Side Information: Incorporating Geographic Relevance into Neighborhood Impact Estimation** [Draft available]
Sole-authored; Journal of Applied Econometrics (2nd revise and resubmit)
2. **The Effects of Order Flow Imbalance on Stock Prices in Tehran Stock Exchange** [Link]
with Mahdi Barakchian; Financial Management Perspective, 2022

WORK IN PROGRESS

1. **Euler Equations for Dynamic Games**
with Victor Aguirregabiria
2. **Who Benefits from Green Subsidies? Labor Standards and the Distribution of Gains**
with Zahra Sedaghat Telgard
3. **One Carbon Market or Many? Optimal Integration of Sectoral Carbon Markets**

HONORS, AWARDS, & FELLOWSHIPS

University of Toronto:	2021–2026
Doctoral Fellowship ($\$20,000 \times 5$); Tuition Fellowship ($\$9,000 \times 5$); Doctoral Recruitment Award ($\$10,000$); Doctoral Completion Award	
Exceptional Talent Scholarship for MSc, Sharif University of Technology	2018–2020
Exceptional Talent Scholarship for BSc, Sharif University of Technology	2014–2018
Silver Medal, 22nd International Mathematics Competition (IMC), Bulgaria	2015
Silver Medal, 55th International Mathematical Olympiad (IMO), South Africa	2014
Bronze Medal, 54th International Mathematical Olympiad (IMO), Colombia	2013

EMPLOYMENT

Susquehanna International Group (SIG)	Summer 2026
Internship offer declined to focus on the academic economics job market.	
University of Toronto — <i>Research Assistant</i>	Summer 2023
To Professor Robert McMillan	
University of Toronto — <i>Teaching Assistant</i>	September 2021–Present
Machine Learning Applications in Economics and Finance (5 semesters)	Undergraduate/Graduate
Financial Economics, Banking, and Corporate Finance (3 semesters)	Undergraduate
Financial Risk Management (3 semesters)	Undergraduate/Graduate
Microeconomics and Microeconomic Theory	Undergraduate/Graduate
Macroeconomics and Global Macroeconomic Policy	Undergraduate
International Trade and Monetary Economics	Undergraduate
Climate Economics	Undergraduate/Graduate
Economic Analysis of Law	Undergraduate
Classical Economic Thought	Undergraduate
Tosan Ofogh Investment Co. , Iran.	Aug 2018–Aug 2021
Co-founded a startup within a hedge fund, conducting market research in the energy sector.	
XaaS Cloud Computing Co. , Iran.	Summer 2017
Software Engineering Intern	
Big Data Lab, Sharif University , Tehran, Iran.	Summer 2016
Research Intern	

ABSTRACTS

When Low-Carbon Incentives Do Not Reduce Emissions: Evidence from Brazil’s RenovaBio

Abstract. This paper studies whether policies that expand low-carbon fuel production necessarily reduce carbon emissions. I examine Brazil’s *RenovaBio* program, a carbon-credit system that provides biofuel producers with incentives funded by fuel distributors. I estimate a unified equilibrium model linking heterogeneous ethanol supply, differentiated downstream fuel demand under oligopolistic pricing, and a dynamic carbon-credit market with banking. The answer is no: RenovaBio increases ethanol production, but because the additional ethanol displaces very little gasoline and instead largely expands total fuel consumption, the policy actually increases CO_2 emissions. In contrast, increasing the ethanol blending mandate by about one percentage point generates essentially the same ethanol production while reducing emissions and creating smaller fuel-price distortions.

Empirical Bayes with Side Information: Incorporating Geographic Relevance into Neighborhood Impact Estimation

Abstract. This paper introduces a novel methodology for estimating neighborhood impacts on intergenerational mobility. The approach employs a nonparametric empirical Bayes framework that addresses geographic heterogeneity in the underlying data-generating process. First, we propose a theoretical framework to address the long-standing yet unresolved question of how to make better use of highly relevant samples—in this context, data from geographically closer regions—when designing a shrinkage estimator for a target area. We demonstrate, both theoretically and empirically, how neglecting this factor can lead to systematically distorted estimates. Second, using publicly available data, we demonstrate that our relevance-based empirical Bayes method reduces Mean Squared Error (MSE) by 26% to 34% relative to benchmark estimates.

The Effects of Order Flow Imbalance on Stock Prices in Tehran Stock Exchange

Abstract. We investigate the effects of order-book events on the prices of the 30 largest stocks on the Tehran Stock Exchange during 2020. We measure the sensitivity of prices to changes in supply and demand volumes and identify the factors affecting this sensitivity. Following the approach of Cont et al. (2014), we conduct approximately 30,000 OLS regressions and show that, in a low-depth market, mid-price returns are significantly explained by order flow imbalance, which captures the net change in demand—that is, the difference between event volumes on the two sides of the order book. We also show that incorporating order flow imbalance from the first three levels of the order book increases the model’s explanatory power for changes in stock mid-prices, indicating that deeper levels of the order book also significantly affect price changes. Moreover, the results confirm that market depth reduces the price impact of order-book events. Our findings are robust across months and stocks.

Who Benefits from Green Subsidies? Labor Standards and the Distribution of Gains

Abstract. How are the gains from green investment subsidies distributed between firms and workers? This project examines whether labor conditions attached to these subsidies increase workers’ share of the benefits, using Canada’s Clean Technology Investment Tax Credit. The policy offers a larger credit to firms meeting prevailing-wage and apprenticeship requirements, linking investment support to employment practices. The analysis links firms’ investment and contractor choices to worker earnings. Firms can qualify for additional support without raising existing workers’ pay if they already meet the standards or hire contractors that do. Higher wages at subsidized projects therefore need not reflect gains caused by the policy. Comparing subsidies with and without labor conditions at the same fiscal cost would assess whether these requirements increase worker earnings and training opportunities, and how they affect investment and hiring.

One Carbon Market or Many? Optimal Integration of Sectoral Carbon Markets

Abstract. This paper asks whether carbon credit markets that jointly penalize dirty activity and reward clean activity should be organized separately by sector or integrated into a common market. Integration can establish a common carbon price, equalizing marginal abatement costs across sectors and shifting emissions reductions toward lower-cost opportunities, while also allowing payments from dirty activity in one sector to finance clean investment in another. Separation, however, preserves sector-specific certificate prices, allowing clean incentives to respond independently to differences in innovation and adoption frictions. I study a market design in which clean activities generate tradable certificates and dirty activities create demand for the same asset, with sector-specific crediting and obligation rates determining the effective clean subsidy and carbon penalty. I characterize when a unified market can achieve the efficient combination of a common carbon price and differentiated clean-technology support, when separate sectoral markets perform better, and how this tradeoff depends on abatement costs and clean-technology spillovers across sectors.